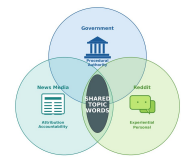
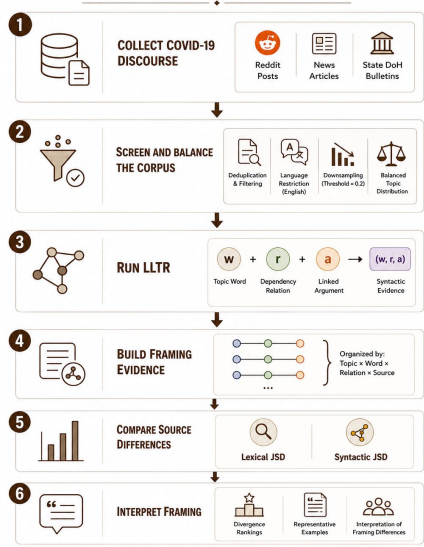


Paper Overview

This paper examines how **COVID-19** was framed differently across **Reddit**, **mainstream news**, and **government communications**. Using the **Linked Latent Theta Role model (LLTR)** as a grammar-aware analytic tool, we compare not only what **topic words** each source used, but also how those words appeared in **sentence structures**.

Across eight COVID-19 topics, we find that sources often shared similar vocabularies while differing substantially in how those words were used in sentence structures. These differences reveal how platforms **assign agency, responsibility, and meaning in distinct ways**. The study contributes a **scalable and interpretable approach** for comparing framing across communication sources beyond keyword-based topic analysis.

ANALYSIS WORKFLOW

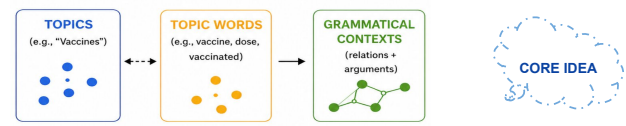


DATA & SOURCE

COVID-19 discourse from March 2020 to December 2022
Reddit: 549 discourse units from r/COVID19 and r/Coronavirus
News: 518 manually screened articles from six U.S. outlets
Government: 570 state Department of Health bulletins from 27 states.
Note: Corpus was source-balanced to support cross-platform comparison.

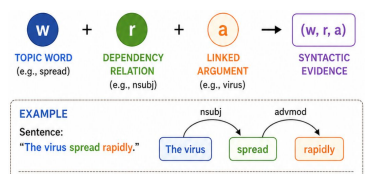
METHOD: LATENT THETA ROLE MODEL (LLTR)

LLTR jointly discovers topics and captures how topic words are grammatically related to other words in sentences.



How LLTR works

For each word occurrence, LLTR links a topic word to the grammatical relation it appears in and the argument it connects to, producing evidence in the form (**Word, Relation, Argument**).



Why LLTR Supports Framing Analysis

The same topic word can be used in different sentence structures. These grammatical differences show how sources **assign agency, responsibility, and meaning**. Because LLTR outputs concrete relation-argument pairs and example sentences, it helps researchers and analysts to inspect and interpret the framing evidence.

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 If you are interested in our work, please scan the QR code to learn more. I would also be happy to connect and discuss it further.



Main Pattern: Similar Words, Different Structures

Figure 2 shows that sources often share similar COVID-19 topic vocabulary. Figure 3 shows stronger syntactic divergence, meaning that the same topic words are framed differently through sentence structure.
Note: Lower JSD = more similar; higher JSD = more different.

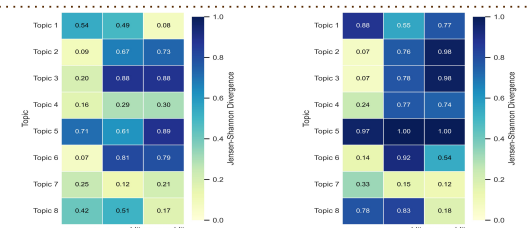


Figure 2. Lexical divergence across sources. Asking: Do sources use similar COVID-19 topic words?

Figure 3. Syntactic divergence across sources. Asking: Do sources use those words in similar sentence structures?

CASE STUDY: TOPIC 5 - POLITICS & PARTISANSHIP

TOPIC CONTEXT

COVID-19 discourse about political actors, partisan conflict, institutional action, and public evaluation.

TOPIC WORDS (Top Keywords)

president administration Biden Trump Congress political

SYNTACTIC DIVERGENCE (JSD)

All source pairs JSD = 0.97 (Very High Divergence)

GOVERNMENT
State DoH Bulletins

political (Topic Word) → pressure (Issue) → not political pressure (Linked Argument)

GRAMMATICAL RELATION (nsbj) LINKED ARGUMENT

EXAMPLE SENTENCE: "We must follow the science, not political pressure, to protect public health."
— State DoH Bulletin

FRAMING IMPLICATION: Politics is framed as a state of conflict and polarization, emphasizing disagreement and institutional decision-making.

NEWS MEDIA
U.S. News Article

political (Topic Word) → issue (Issue) → divisive political issue (Linked Argument)

GRAMMATICAL RELATION (nsbj) LINKED ARGUMENT

EXAMPLE SENTENCE: "The stimulus package has become a divisive political issue in Congress."
— U.S. News Article

FRAMING IMPLICATION: Politics is framed as a state of conflict and polarization, emphasizing disagreement and accountability.

REDDIT
r/COVID19 & r/Coronavirus

political (Topic Word) → theater / games (Issue) → political theater (Linked Argument)

GRAMMATICAL RELATION (nsbj) LINKED ARGUMENT

EXAMPLE SENTENCE: "It's all political theater—nobody actually cares about people."
— Reddit Comment

FRAMING IMPLICATION: Politics is framed as a performance and optical, characterized by gamesmanship and perceived irresponsibility.

SUMMARY & NEXT STEP

Shared vocabulary does not mean shared framing. Our analysis shows that Reddit, news media, and government communications often used similar COVID-19 topic words, but structured them differently in sentences. These grammatical differences reveal how sources assign agency, responsibility, and meaning. Future work can apply this grammar-aware framing approach to other crises, political events, and multilingual settings, while adding human validation and richer semantic role analysis.

Acknowledgements

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Paper Overview

Our research examines how **COVID-19** was framed differently across **Reddit**, **mainstream news media**, and **government communications**.

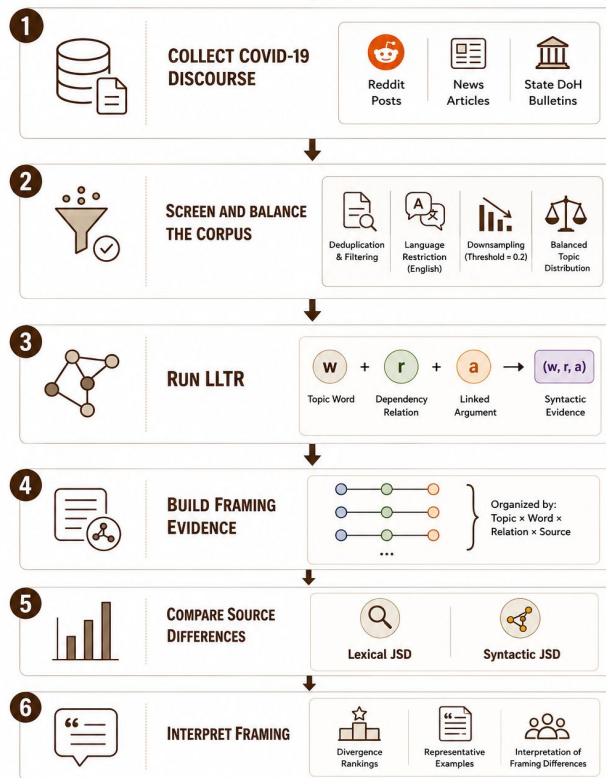
By framing, we mean how a source shapes people's understanding of an issue.

Using the **Linked Latent Theta Role model (LLTR)** as a grammar-aware analytic tool, we compare not only what **topic words** each source used, but also how those words appeared in **sentence structures**.

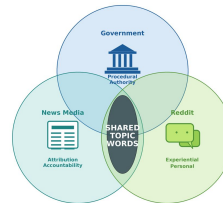
These differences reveal how platforms **assign agency, responsibility, and meaning** in distinct ways.

The study contributes a **scalable and interpretable approach** for comparing framing across communication sources beyond keyword-based topic analysis.

ANALYSIS WORKFLOW



Key Takeaway: Across eight COVID-19 topics, we find that sources often shared similar vocabularies while differing substantially in how those words were used in sentence structures.



Government: The Procedural Frame
Emphasizes institutional authority, eligibility, and compliance.



News Media: The Attribution Frame
Focuses on expert mediation, conflict, and reporting accountability.



Reddit: The Experiential Frame
Centers on personal judgment, lived experience, and evaluative stances.

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